A Contrastive Learning Enhanced Adaptive Multimodal Fusion Network for Hyperspectral and LiDAR Data Classification

Kai Xu, *Member, IEEE*, Bangjun Wang, Zhou Zhu, Zhaohong Jia, and Chengcheng Fan

***Abstract*— In recent years, the fusion-based classification of hyperspectral image (HSI) and light detection and ranging (LiDAR) data has garnered increasing attention from researchers. However, significant image disparities exist between HSI and LiDAR data because of their distinct imaging mechanisms, which limits the fusion of HSI and LiDAR data. Therefore, establishing effective interaction (INTER) and fusion between HSI and LiDAR data is crucial. Moreover, the classification of multimodal data with limited labeled instances is another challenging task. To address these issues, we propose a contrastive learning enhanced adaptive multimodal fusion network (CAMFNet) for the joint classification of HSI and LiDAR data. CAMFNet introduces a novel semantic similarity contrastive (SSC) loss that fully utilizes a large amount of unlabeled data to learn potential complementary information between different modalities. In addition, a hierarchical fusion strategy is adopted to fuse multimodal features. At the shallow feature fusion stage, a cross-modal feature interaction fusion (CMFIF) module is proposed to guide the extraction of multimodal features and enhance the complementary information through cross-modal interaction. At the same time, adaptive feature fusion is also used to dynamically assign corresponding weights to different modal features, achieving finer fusion. At the deep feature fusion stage, a channel split and integration (CSI) block is constructed to further optimize the fused spectral–spatial and elevation features via a negative feedback mechanism. The effectiveness of CAMFNet is validated through experiments on three standard datasets, which demonstrates that CAMFNet outperforms state-of-the-art methods.**

***Index Terms*— Contrastive learning, cross-modal interaction (INTER), feature fusion, hyperspectral image (HSI), image classification, light detection and ranging (LiDAR).**

I. Introduction

REMOTE sensing (RS) imagery classification plays a crucial role in environmental monitoring and protection, urban planning and management, and disaster monitoring by providing essential information [1], [2], [3], [4], [5]. In early research, numerous methods for hyperspectral image (HSI) classification were proposed [6], [7], [8], [9]. However, due to atmospheric interference during HSI acquisition, there may be instances where the same object exhibits varying spectral responses, and different objects may exhibit similar spectral responses [10], [11]. In addition, the limited spatial resolution of HSI poses challenges for fine-grained classification. With advancements in RS imaging technology, different modal data describing the same area can be obtained via various sensors. For example, light detection and ranging (LiDAR) can provide depth and intensity information about surface cover materials [12]. It helps to distinguish between objects with similar spectral characteristics but varying elevation information, such as roads and rooftops made of concrete. Consequently, research into image classification methods based on the fusion of HSI and LiDAR information has emerged as a focal point of study [13], [14].

Collaborative classification based on HSI and LiDAR data aims to effectively utilize multimodal data by comprehensively fusing the spectral–spatial information of HSI and the elevation information of LiDAR [15], [16], [17]. However, due to differences in imaging mechanisms, HSI and LiDAR data often exhibit significant image disparities, rendering methods relying on cascade or pixelwise feature fusion ineffective. To achieve accurate classification results, existing methods largely focus on strategies such as exploring multiscale fusion and attention mechanisms [18], [19], [20], [21]. These methods aim to extract and integrate similar higher level semantic information and complementary discriminative information from multimodal data. Although these methods consider modal differences, they overlook interaction (INTER) between features extracted from multiple modalities during the feature extraction process [22], [23], [24]. Given the complexity of RS image classification, feature interaction can fully capture correlations between multimodal features, enrich subsequent features, and enhance classifier discriminative ability [25]. Some methods enhance information interaction between different modalities through shared feature extractors or weighted auxiliary modalities, but they do not explicitly exploit the complementary properties of different modalities to enhance each other, i.e., leveraging the strengths of one modality to supplement another [26], [27], [28].

Moreover, current methods rely heavily on a large amount of labeled data for enhancing performance, but acquiring labeled data for RS images is time-consuming and requires

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specialized knowledge [29], [30], [31], [32]. Therefore, the rise of unsupervised learning is driven [33]. Unlike supervised learning, unsupervised learning extracts useful information by learning the inherent structure and distribution of data [34]. Existing unsupervised learning methods utilize unlabeled data through multimodal RS image fusion and then reconstruction [35], [36], [37], [38]. However, due to inconsistent feature representations between different modalities, information loss or confusion occurs during the multimodal fusion process, leading to relatively low prediction accuracy and interpretability of the final reconstruction results.

To address the aforementioned issues, we propose a contrastive learning enhanced adaptive multimodal fusion network (CAMFNet) that combines HSI and LiDAR via a dual-branch convolutional neural network (CNN) with hierarchical fusion strategies to achieve efficient multimodal data fusion. Specifically, a novel contrastive loss known as semantic similarity contrastive (SSC) loss is designed to learn multimodal features from unlabeled data. Building upon this, the entire network adopts a hierarchical fusion strategy for integrating multimodal features. At the shallow feature fusion stage, the cross-modal feature interaction fusion (CMFIF) module is employed to guide the extraction of multimodal features, ensuring the retention of relevant and complementary features while reducing redundancy. The CMFIF module also adaptively merges multimodal features, dynamically assigning appropriate weights to different modal features. At the deep feature fusion stage, the channel split and integration (CSI) block is developed to integrate information from different channels of the obtained modal fusion features, achieving the refusion of multimodal fusion features.

The primary contributions can be summarized as follows.

1) To further optimize the fusion of spectral–spatial and elevation features, we design the CMFIF module. Considering the correlations and complementarities between different modalities, it facilitates more refined feature fusion.
2) To explore the possibility of deep refusion of multimodal fusion features, we propose the CSI block to integrate diverse channel information.
3) To address the costly labeling issue, we introduce a contrastive learning strategy. By employing SSC loss, we extract potential complementary information between different modalities, achieving cross-modal semantic alignment.
4) Extensive experiments are carried out on several RS datasets, and the experimental results show that CAMFNet surpasses other existing methods on benchmark datasets.

The organization of the remaining sections is given as follows. Section II offers a concise overview of related work on multimodal data fusion classification and contrastive learning. Section III details the proposed network architecture. Section IV covers the experimental data, results, and analysis. Finally, Section V presents conclusions.

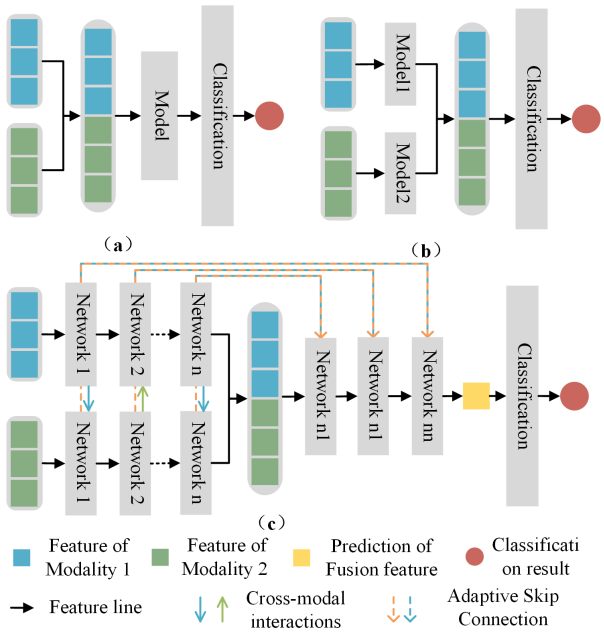


Fig. 1. Existing fusion structures. (a) Early fusion. (b) Late fusion. (c) Dense fusion.

II. Related Work

A. Multimodal Data Fusion Classification

With the rapid advancement of RS technology, the application of multimodal fusion strategies in the field of RS scene classification has become increasingly widespread. Its principal advantage lies in the ability to integrate unique information from different sensors, significantly enhancing the richness and accuracy of feature extraction. Currently, multimodal fusion methods are primarily developed in three dimensions: the pixel level, the feature level, and the decision level [39], [40], [41]. Among these methods, feature-level fusion is particularly advantageous for its effectiveness in managing heterogeneous data such as HSI and LiDAR data.

Current research predominantly focuses on single-level feature fusion strategies, which implement early or late fusion at specific layers of neural networks with the objective of integrating integrated features extracted from different modalities. For example, dual attention based spectrospatial multimodal fusion network (FusAtNet) [42] innovatively integrates HSI and LiDAR data early in the data processing stage via self-attention and cross-attention mechanisms, generating rich attentional representations in joint spectral–spatial information, as illustrated in Fig. 1(a). This significantly enhances land cover classification performance. Furthermore, deep encoder–decoder networks (EndNet) proposed by Hong et al. [43] adopts a deep encoder–decoder architecture and employs a late fusion strategy to reconstruct multimodal inputs, effectively activating cross-modal neural connections and demonstrating superior fusion results compared with traditional methods, as shown in Fig. 1(b).

Although these methods have achieved notable advancements in multimodal RS image classification, they still face several challenges that urgently need to be addressed. Specif-

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ically, early fusion strategies can amplify noise and errors due to inherent differences between modalities, making it difficult to comprehensively capture complex correlations and heterogeneities between different modalities. In contrast, while late fusion strategies can alleviate some of these issues to a certain extent, they may fail to consider the dynamic interaction between different modalities, which could result in the overlooking of crucial information and even amplifying errors from early-stage decisions.

To address the shortcomings of current methods, we propose an innovative multilevel hierarchical fusion structure. As depicted in Fig. 1(c), CAMFNet is carefully designed as a multilevel and multiscale feature fusion structure. Specifically, CAMFNet employs a multimodal interaction to facilitate the initial integration of information across different modalities at the shallow feature fusion stage. Subsequently, at the deep feature fusion stage, the channel information of the multimodal fusion features is integrated further to facilitate the exploration and utilization of cross-modal deep-level information in a more comprehensive and accurate manner.

B. Contrastive Learning

Contrastive learning, gaining traction in natural image processing, requires minimal data annotation efforts [44], [45]. Its essence lies in designing specific contrastive tasks to distinguish positive pairs from negative pairs, thereby automatically extracting and optimizing feature representations without relying on label information. In recent years, this strategy has demonstrated significant efficacy in the field of HSI classification, notably through pioneering work by Hang et al. [46]. They constructed a cross-modal contrastive loss function that effectively explores semantic and structural information between HSI and LiDAR data. Through a dual fine-tuning strategy, they successfully transferred features from limited samples to the reused branch of the HSI, resulting in a marked improvement in HSI classification performance.

As contrastive learning methods advance quickly, researchers are increasingly applying them to more complex multimodal RS data processing [47], [48]. To comprehensively capture intramodal feature representations and explore potential correspondences across multimodal data in detail, Feng et al. [49] proposed a novel framework called cross-modal contrastive learning under a self-supervised setting. The framework integrates intramodal contrastive learning with cross-modal contrastive learning, aiming to efficiently mine the feature potentials of multimodal data during pretraining. This ensures that the learned feature representations maintain high semantic consistency both intramodal and cross-modal.

However, it is important to highlight that existing contrastive learning methods for processing multimodal RS data typically generate positive and negative samples through data augmentation. These approaches often fail to fully exploit the spatial distribution characteristics between positive and negative samples, as well as the semantic relationships across neighboring instances [50]. This limitation restricts the model's generalization ability in complex scenarios.

To address this problem, we innovatively propose a contrastive learning strategy that integrates mean square error (MSE) loss with SSC loss. Specifically, MSE loss facilitates mutual learning and alignment of feature maps across different modalities, but it is inadequate for maximizing differences between negative samples. To complement the MSE loss, we develop a contrastive loss based on Dice similarity, which we designate as SSC loss. This strategy optimizes multimodal feature representations through interregion mutual learning directly, without relying on the ground truth, thereby significantly leveraging the spatial distribution characteristics and semantic similarity of neighborhoods.

III. Methodology

A. Overall Architecture

To reduce reliance on labeled data and enhance fusion quality, CAMFNet is designed. As depicted in Fig. 2, CAMFNet consists of three main components: the CMFIF module, the CSI block, and the classification block. In particular, CMFIF includes spectral-to-elevation feature interaction (SEFI), elevation-to-spectral feature interaction (ESFI), and adaptive modality fusion (AMF). Next, we detail the specific implementation process of the network.

The CAMFNet training process involves two major steps. First, we employ a contrastive learning strategy based on MSE loss and SSC loss, as shown in Fig. 3(a). This strategy leverages the complementary properties of multimodal unlabeled data to improve the capability of the end-to-end network, thereby acquiring a pretrained encoder–decoder feature extraction network that improves feature representation quality. Subsequently, the CSI block is integrated into the pretrained feature extraction network, and fine-tuning is performed using a limited set of labeled data, as depicted in Fig. 3(b). The CSI block further optimizes the fusion feature of HSI and LiDAR data, ensuring the integrity and consistency of information at the channel level, thereby enhancing the model's classification accuracy and generalization capability in complex environments. Finally, the integrated features are processed through the classification block to obtain precise pixelwise classification results. It is noteworthy that the CMFIF module is involved in two distinct stages of the training process.

B. CMFIF Module

In the process of multimodal data fusion, effectively utilizing interaction between different data modalities is crucial for data representation capabilities and improving classification accuracy. To exploit supplementary information between HSI and LiDAR data to improve their respective feature representations, we design the CMFIF module at the shallow feature fusion stage, that is, SEFI, ESFI, and AMF.

1) Feature Interaction: In order to exploit the complementary properties of different modalities, we develop SEFI and ESFI to guide the extraction of multimodal features. The specifics of the SEFI are illustrated in Fig. 4(a). SEFI uses linear layers to generate channel attention weights of size $C \times C$, followed by matrix multiplication and cross-addition operations, thereby extracting the dominant complementary features in the spectral–spatial features related to elevation features. The dominant complementary feature is used to enhance

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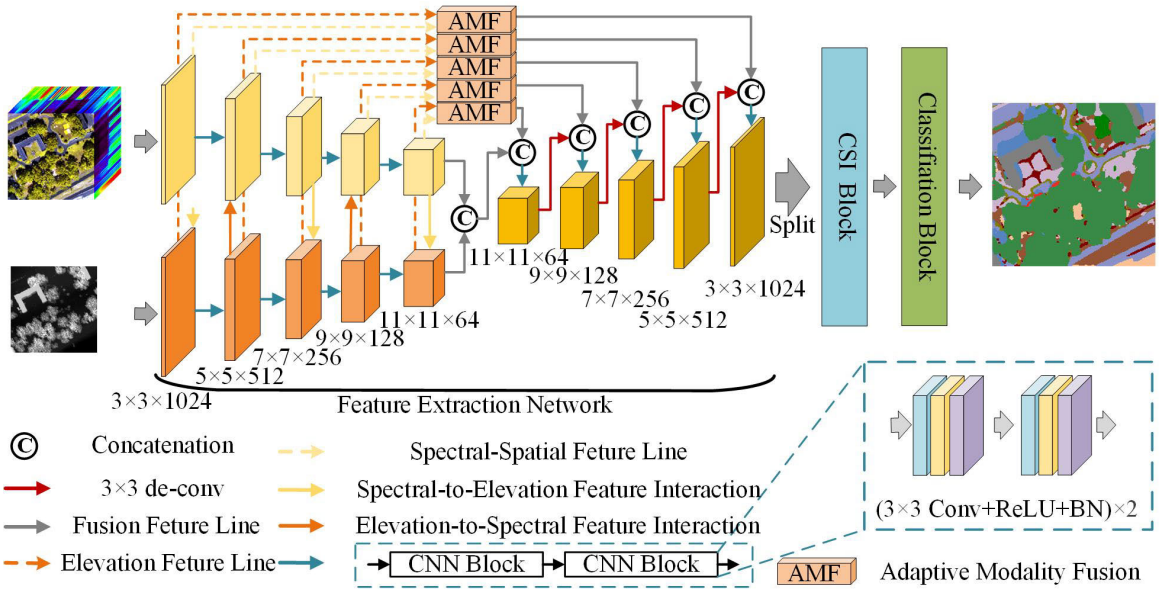


Fig. 2. Overall network architecture of CAMFNet. Include the CMFIF (i.e., SEFI, ESFI, and AMF) module in the feature extraction network, and then, the multimodal fusion features are refused by the CSI block and, finally, fed to the classification block.

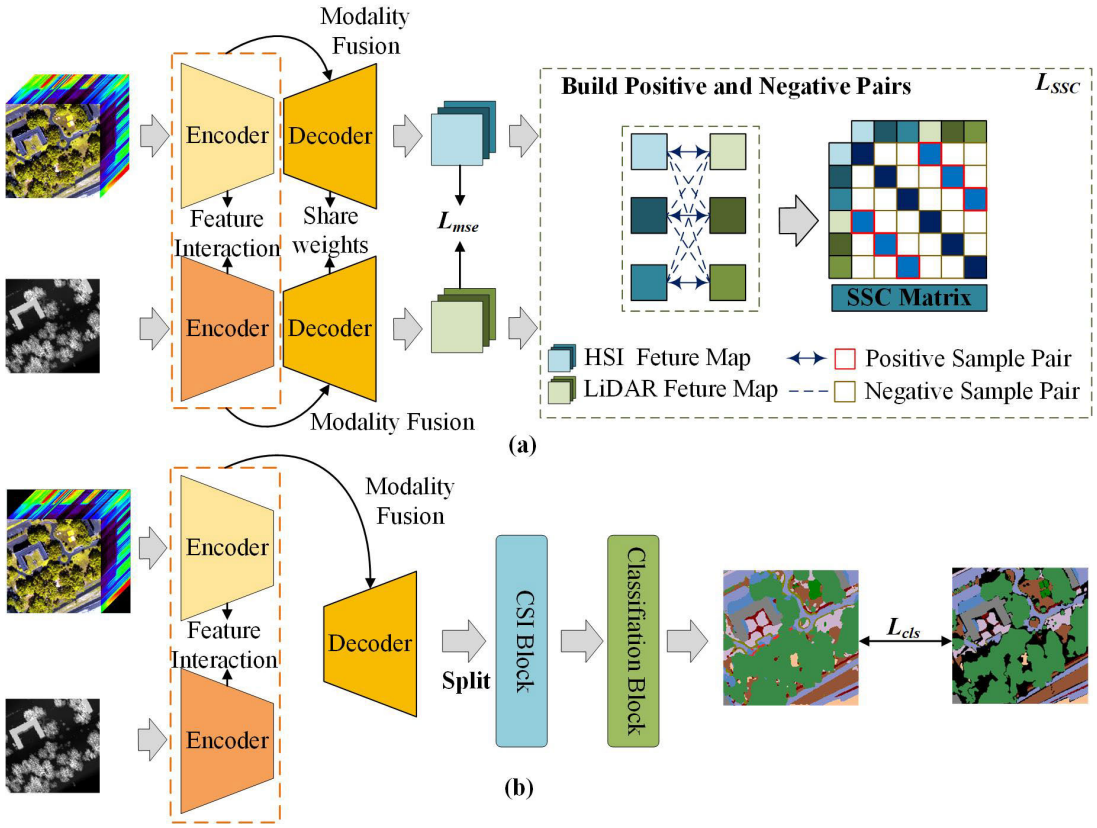


Fig. 3. Contrastive learning enhanced adaptive multimodal fusion network. (a) Contrastive learning based on SSC loss. (b) Classification fine-tuning learning.

the elevation features and guide the subsequent extraction of elevation features. The SEFI can be given as

$$\hat{f}_{iH} = f_{iH} \oplus LL_v\left(\left[f_{iH}, f_{iL}\right]\right) \otimes LL_q(f_{iL}) \otimes LL_k(f_{iH}) \quad (1)$$

$$\hat{f}_{iL} = f_{iL} \oplus LL_v\left(\left[f_{iH}, f_{iL}\right]\right) \otimes LL_q(f_{iH}) \otimes LL_k(f_{iL}) \quad (2)$$

$$\tilde{f}_{iL} = f_{iL} \oplus \mathrm{conv}_3\left(\mathrm{concat}\left(\hat{f}_{iH}, \hat{f}_{iL}\right)\right) \quad (3)$$

where [,] represents concatenation operations, LL_q, LL_k, and LL_v represent different linear operations, and f_{iH} and f_{iL} denote the spectral and elevation features that are produced by the convolution block of encoder layer i ($i = 1, 3, 5$), respectively. $\hat{f}_{iH}$ and $\hat{f}_{iL}$ denote channel interaction features, conv_3 represents a 3×3 convolution operation, $\tilde{f}_{iL}$ represents the elevation features after interaction for the ith layer, and the

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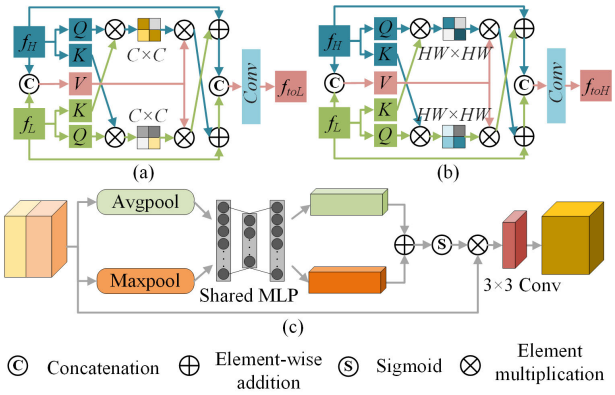


Fig. 4. Details of the CMFIF module. (a) SEFI. (b) ESFI. (c) AMF.

operations $\oplus$ and $\otimes$ denote the elementwise addition and the matrix multiplication, respectively.

The details of the ESFI are depicted in Fig. 4(b). ESFI operates similar to SEFI by producing spatial attention weights of size $HW \times HW$, which are used to extract key complementary features from the elevation features that are associated with the spectral–spatial features. As a result, ESFI enhances the spectral–spatial features and guides the extraction of the spectral–spatial feature in subsequent processes. The ESFI can be formulated as

$$\hat{f}_{iH} = f_{iH} \oplus LL_v\left(\left[f_{iH}, f_{iL}\right]\right) \otimes LL_q(f_{iL}) \otimes LL_k(f_{iH}) \quad (4)$$

$$\hat{f}_{iL} = f_{iL} \oplus LL_v\left(\left[f_{iH}, f_{iL}\right]\right) \otimes LL_q(f_{iH}) \otimes LL_k(f_{iL}) \quad (5)$$

$$\bar{f}_{iH} = f_{iH} \oplus \text{conv}_3\left(\text{concat}\left(\hat{f}_{iH}, \hat{f}_{iL}\right)\right) \quad (6)$$

where f_{iH} and f_{iL} denote the spectral and elevation features output by the convolutional block of encoder layer $i(i = 2, 4)$, respectively. $\hat{f}_{iH}$ and $\hat{f}_{iL}$ denote spatial interaction features, and $\bar{f}_{iH}$ represents the spectral–spatial features after interaction for the ith layer.

The SEFI and ESFI enhance the correlation and complementarity between features of different modalities, laying the foundation for subsequent fusion processes through mutual guidance of complementary features.

2) Adaptive Modality Fusion: To minimize redundancy in modal fusion features while preserving their richness and information integrity, AMF is proposed. Details of the AMF are depicted in Fig. 4(c).

During the AMF of the CMFIF module, global average pooling p_{avg} and global maximum pooling $p_{\max}$ are employed to capture global channel information from concatenated feature maps. The global channel information is used to automatically highlight relevant feature channels while suppressing irrelevant. Subsequently, the channel attention coefficients A_{avg} and $A_{\max}$ are generated by shared multiple-layer perceptron, which comprises two fully connected layers and an ReLU layer

$$A_{\text{avg}} = M\left(p_{\text{avg}}\left(\left[f_{1H}, \tilde{f}_{1L}\right]\right)\right) \quad (7)$$

$$A_{\max} = M\left(p_{\max}\left(\left[f_{1H}, \tilde{f}_{1L}\right]\right)\right) \quad (8)$$

where f_{1H} and $\tilde{f}_{iL}$ represent the feature outputs after interaction by the convolutional block of the first layer of the encoder. M denotes a shared multiple layer perception layer, where the first fully connected layer has $C/2$ output channels to balance performance and computational cost tradeoffs. Finally, a sigmoid function is used to generate the global channel attention map A, which weights the concatenated feature maps. The specific operation can be formulated as

$$A = s\left(A_{\text{avg}} \oplus A_{\max}\right) \quad (9)$$

$$f_{1\text{fuse}} = \text{Conv}_3\left(A \odot \left[f_{1H}, \tilde{f}_{1L}\right]\right) \quad (10)$$

where s represents the sigmoid function, $f_{1\text{fuse}}$ denotes the final fused modality feature map, and $\odot$ signifies the elementwise multiplication.

The AMF operates similarly in other layers of the encoder by adaptively weighting each modality according to its importance to reduce information redundancy. Moreover, the AMF establishes connections between the encoder and decoder, transmitting shallow encoder features to the corresponding deep decoder parts. This effectively prevents the loss of shallow features during fusion, thereby preserving feature richness and information integrity.

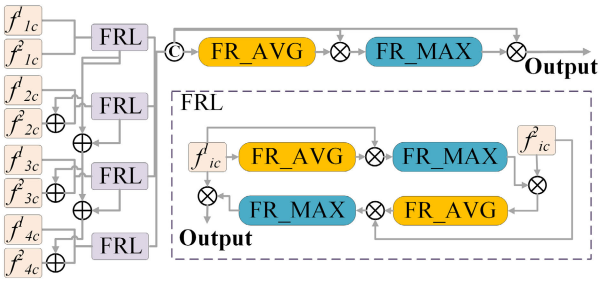


Fig. 5. Illustration of the CSI block.

C. CSI Block

Considering the unevenness and complementarity of information across different channels, the CSI block is proposed in the deep feature fusion strategy to explore the refusion of multimodal fusion features. The CSI block aims to enhance the model's capacity to capture complex relationships and patterns in multimodal data, as demonstrated in Fig. 5.

Unlike simple elementwise addition or multiplication, the CSI block employs a novel feedback–negative feedback learning mechanism termed feature reintegration learning (FRL). FRL is designed to ensure comprehensive and consistent information at the channel level. Specifically, after fusing features from two modalities, the resulting feature maps along the channel dimension are split to obtain two equally sized maps, f_c^1 and f_c^2. Subsequently, each split portion is further subdivided into four parts, followed by channel information integration and processing via FRL.

The realization of FRL primarily involves two operations: FR_AVG and FR_MAX, as illustrated in Fig. 6. In FRL, the initial step is to apply 1-D global average pooling (1-D-GAP) separately along the width and height dimensions. Unlike previous attention mechanisms that utilize 2-D global pooling, which can lead to loss of positional information,

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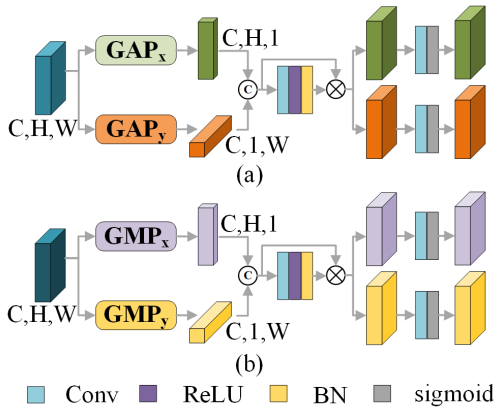


Fig. 6. Details of (a) FR_AVG and (b) FR_MAX.

1-D-GAP offers the advantage of preserving spatial dimensions in one direction while maintaining a large receptive field. One-dimensional GAP allows for the precise capture of the spatial location information of ground objects. The specific operation is given as follows:

$$f_h = \frac{1}{W} \sum_{0 \le i \le W} f_{ic}^1(H, i) \tag{11}$$

$$f_w = \frac{1}{H} \sum_{0 \le i \le H} f_{ic}^1(i, W) \tag{12}$$

where W and H denote the width and height of the split feature maps, respectively, with $f_{ic}^1, i \in \{1, 2, 3, 4\}$ representing one of the feature maps segmented from the feature map f_c^1 along the channel dimension. f_h and f_w are aggregated features in two spatial directions. The subsequent convolutional layers adaptively adjust the characteristic response values of each channel. The specific operations are detailed as follows:

$$f_{hw} = \mathrm{BN}\big(\mathrm{conv}_1\big(\big[f_h, f_w\big]\big) \otimes \sigma\big(\mathrm{BN}\big(\mathrm{conv}_1\big(\big[f_h, f_w\big]\big)\big)\big) \tag{13}$$

where conv_1 represents a 1×1 convolutional layer and σ signifies the ReLU activation function. BN stands for batch normalization. In addition, to reduce the computational effort, a reduction ratio of 2 is introduced in the convolutional layer.

Finally, the obtained feature maps are separated along their width and height dimensions. To prevent the loss of shallow layer information, skip connections are employed to supplement these shallow features. The FR_AVG operations are formalized as follows:

$$\hat{f}_{ih}, \hat{f}_{iw} = s(\mathrm{conv}_1(\mathrm{SP}(f_{hw}))) \tag{14}$$

$$f_{i\mathrm{avg}}^1 = f_{ic}^1 \odot \hat{f}_{ih} \odot \hat{f}_{iw} \tag{15}$$

where SP signifies the channelwise splitting operation. $f_{i\mathrm{avg}}^1$ is the output of the action of FR_AVG. Subsequently, a similar operation is executed for f_{iavg}^1, with the exception that the 1-D-GAP is substituted with 1-D maximum pooling. The FR_MAX operations are formalized as follows:

$$f_{i\max}^1 = f_{ic}^2 \odot \hat{f}_{ih} \odot \hat{f}_{iw} \tag{16}$$

where $f_{i\max}^1$ is the output of the action of FR_MAX.

Consequently, the FR_AVG and FR_MAX implement feedback processing of one set of segmented multimodal fusion feature maps f_{ic}^2 through another set of segmented multimodal fusion feature maps f_{ic}^1. Similarly, FR_AVG and FR_MAX are also applied to the acquired another set of segmented feature maps, constituting a feedback–negative feedback processing sequence

$$f_{i\max}^2 = \mathrm{FR}\big(f_{ic}^2\big) \tag{17}$$

where FR is composed of both FR_AVG and FR_MAX.

The final output f_{csi} of the CSI block can be formulated as

$$f_{\mathrm{csi}} = \mathrm{FR}\big(\big[f_{1\max}^2, f_{2\max}^2, f_{3\max}^2, f_{4\max}^2\big]\big). \tag{18}$$

D. Network Training Strategy

1) Contrastive Learning Based on SSC Loss: In RS images, adjacent regions often share the same labels. Inspired by the correlation of multimodal data in this study, we employ MSE loss and SSC loss with the objective of leveraging the semantic similarity between neighboring regions. This contrastive learning strategy enhances cross-modal semantic alignment and utilizes the characteristics of the spatial distribution of neighborhoods. Specifically, we treat identical regions across different modalities as positive sample pairs and establish negative sample pairs both intramodal and intermodal to explore cooperative and differential information within and between modalities. We define the unlabeled datasets ζ^+ and ζ^- as follows:

$$\zeta^+ = \big\{\big(x_H^i, x_L^i\big)\big\}_{i=1}^n, \quad \zeta^- = \Big\{\Big(x_H^j\Big)\Big\}_{j \ne i}^{n-1} \cup \Big\{\Big(x_L^j\Big)\Big\}_{j \ne i}^{n-1} \tag{19}$$

where x_H^i, x_L^i, x_H^j, and x_L^j are different image modalities. Positive and negative samples are batched into the network, proceeding through the encoder to obtain the spectral–spatial feature y_H^i and elevation feature y_L^i. Following this, y_H^i and y_L^i are, respectively, concatenated with the output of the AMF module and fed into the decoder to obtain spectral–spatial feature ℓ_h^{out} and elevation feature ℓ_l^{out}. In addition, the SEFI and ESFI affect the entire encoder section.

Next, we present the specific implementation for the proposed SSC loss. Initially, to measure the contextual semantic similarity between pairs of positive and negative samples, we employ the Dice coefficient similarity as a metric function, which is defined as follows:

$$S_{\mathrm{dice}}(\alpha_1, \alpha_2) = \frac{2 \times \sum_{\substack{\hat{\alpha}_1 \in \alpha_1, \\ \hat{\alpha}_2 \in \alpha_2,}} \hat{\alpha}_1 \hat{\alpha}_2}{\sum_{\substack{\hat{\alpha}_1 \in \alpha_1, \\ \hat{\alpha}_2 \in \alpha_2,}} (\hat{\alpha}_1 + \hat{\alpha}_2)} \tag{20}$$

where $\hat{\alpha}_1 \hat{\alpha}_2$ denotes the elementwise multiplication and $\hat{\alpha}_1$ and $\hat{\alpha}_2$ denote the values of features α_1 and α_2 extracted from each pixel, respectively. We define the SSC loss for a positive pair as

$$l_{\mathrm{ssc}}\big(x', x''\big) = -\log \frac{\exp\big(S_{\mathrm{dice}}\big(x', x''\big)\big)}{\exp(S_{\mathrm{dice}}(x', x'')) + \sum_{y \in \zeta^-} \exp(S_{\mathrm{dice}}(x', y))} \tag{21}$$

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where (x', x'') refers to the decoder output for positive sample pairs from ζ^+ and y denotes the decoder output for negative samples corresponding to x' from ζ^-. The complete SSC loss for a minibatch containing n unlabeled images can be articulated as follows:

$$\min_{S,I} L_{\mathrm{SSC}}(S, I) = E_{x',x''}\left[\frac{1}{2n}\sum_{(x',x'')\in\zeta^+}\left(l_{\mathrm{ssc}}\left(S\left(x'\right), I\left(x''\right)\right)+l_{\mathrm{ssc}}\left(S\left(x''\right), I\left(x'\right)\right)\right)\right] \tag{22}$$

where $S(\cdot)$ and $I(\cdot)$ signify distinct feature extraction branches, whereas x' and x'' correspond to data from different modalities, and $E_{x',x''}$ denotes averaging. Each sample within a positive pair requires the computation of similarity against samples from the negative samples; thus, positive pairs need to be computed twice l_{ssc}.

Furthermore, a weighting coefficient w is introduced with the objective of harmonizing the MSE loss and SSC loss. The comprehensive loss definition for the first stage training of the CAMFNet is outlined as follows:

$$\min_{S,I} L_{\mathrm{all}}(S, I) = wL_{\mathrm{mse}} + L_{\mathrm{SSC}} \tag{23}$$

where L_{mse} is the MSE loss.

2) Classification Fine-Tuning Learning: To achieve high classification accuracy, we further fine-tune the model with labeled data. Specifically, the CSI block is integrated into the pretrained feature extraction network. In addition, to improve the fusion of HSI and LiDAR data features, we concatenate the spectral–spatial features and elevation features extracted by the encoder and then feed them into the decoder to get the output features ℓ_{hl}^{out}. After processing by the CSI block, the final features are forwarded to the classification block for prediction. During the fine-tuning stage, we utilized the cross-entropy loss function, formulated as follows:

$$L_{\mathrm{cls}} = -\frac{1}{M}\sum_{i=1}^{M} q\log(p) + (1-q)\log(1-p) \tag{24}$$

where M represents the batch size, p signifies the predicted class probability for the ith sample, and q denotes the class label of the ith input data.

The complete training procedure of CAMFNet is illustrated in Algorithm 1.

Algorithm 1 Contrastive Learning Enhanced Adaptive Multimodal Fusion Network

Input: Hyperspectral image **H**, LiDAR image **L**, ground truth **Y**, patch size s = 11.
Output: Predicted labels of the test set.

```
Set the first stage epochs number p1, the second stage epochs number p2, the number of PCA q to 30.
Create all sample patches from H to L, obtain unlabeled dataset Hu, Lu and labeled dataset Hl, Ll.
Generate training loader and test loader.
for i = 1 to p1 do
    Use encoders and decoders to extract the features ℓ_h^out from Hu, and ℓ_l^out from Lu, as described by (1)-(10).
    Get pre-trained encoders and decoders.
end for
Load the pre-trained weights.
for i = 1 to p2 do
    Use the pre-trained encoders and decoders to extract the features l_hl^out from Hl and Ll.
    Acquire the label corresponding to the present pixel using the CSI block and the classification module, as described by (11)-(18).
end for
Utilize the test set alongside the trained model to obtain predicted labels.
```

IV. Experiments and Analysis

In order to evaluate the effectiveness of CAMFNet, three publicly available multimodal RS datasets are utilized in the experiments conducted. An Section IV-A provides a comprehensive description of these datasets, followed by a detailed account of the experimental setup and the presentation of results. These results highlight CAMFNet's effectiveness and its advantages over current methods. Finally, ablation studies are performed to demonstrate the impact of different modules within the CAMFNet.

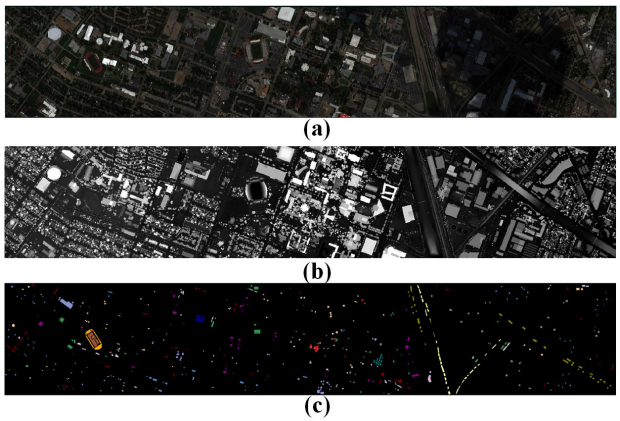


Fig. 7. Houston2013 dataset. (a) Pseudocolor composite image based on bands 59, 40, and 23 for HSIs. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

A. Data Descriptions

To gauge the performance of the CAMFNet, the study uses the Houston2013, Trento, and Michigan Urban Ultra-Fine Lidar Gulfport (MUUFL) datasets for both quantitative and qualitative evaluations.

1) Houston2013 Dataset: Originating from the 2013 IEEE GRSS Data Fusion Contest, this dataset encompasses the University of Houston campus and its neighboring urban areas. Collected in 2012 by the National Center for Airborne Laser Mapping, it includes LiDAR data with a single band. The images for both modalities are 349 × 1905 pixels in resolution. The hyperspectral imagery contains 144 spectral bands ranging from 0.38 to 1.05 μm and includes 15 urban land cover classes, as detailed in Table I. Moreover, Fig. 7 shows a pseudocolor composite of the HSI, a gray-scale LiDAR image, and the ground truth.

2) Trento Dataset: This dataset, obtained from rural southern Trento, Italy, features hyperspectral data captured by the

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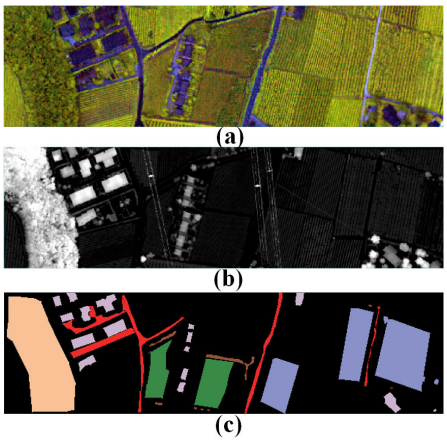


Fig. 8. Trento dataset. (a) Pseudocolor composite image based on bands 59, 40, and 23 for HSIs. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

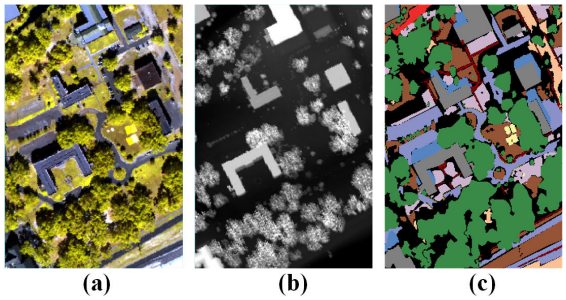


Fig. 9. MUUFL dataset. (a) Pseudocolor composite image based on bands 59, 40, and 23 for HSIs. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

AISA Eagle system with 63 spectral bands (0.42–0.99 μm) and LiDAR data from the Optech ALTM 3100EA sensor. It consists of 600 $\times$ 166 pixels with a spatial resolution of 1 m, including six ground object categories, as listed in Table II. Furthermore, Fig. 8 presents a pseudocolor composite of the HSI data, a gray-scale LiDAR image, and the ground-truth map.

3) MUUFL Dataset: Collected over the University of Southern Mississippi Gulf Park Campus on November 8, 2010, this dataset includes coregistered hyperspectral and LiDAR data with a spatial resolution of 1 m. The hyperspectral data, acquired with the CASI-1500 sensor, feature 64 spectral bands. Both the hyperspectral and LiDAR images are 325 $\times$ 220 pixels in size and include 11 land cover categories, as specified in Table III. In addition, Fig. 9 presents a pseudocolor image of the hyperspectral data alongside a gray-scale LiDAR image.

4) Augsburg Dataset: Captured in Augsburg, Germany, this dataset comprises satellite-based HSI and digital surface model (DSM)-based LiDAR data. Specifically, the HSI and LiDAR data were acquired by Data Assimilation System for the Earth Observation Centre (DAS-EOC) and German Aerospace Center 3K Hyperspectral Imaging System (DLR-3K). Upon reviewing the original dataset, we observe that the labeling is based on functional zoning principles. However, a significant issue arises from noticeable overlaps between the label categories. For instance, certain areas are labeled as "Residential,"

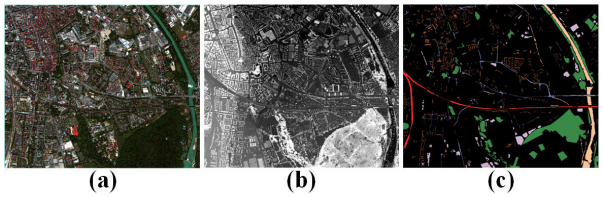


Fig. 10. Augsburg dataset. (a) Pseudocolor composite image based on bands 50, 23, and 8 for HSIs. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

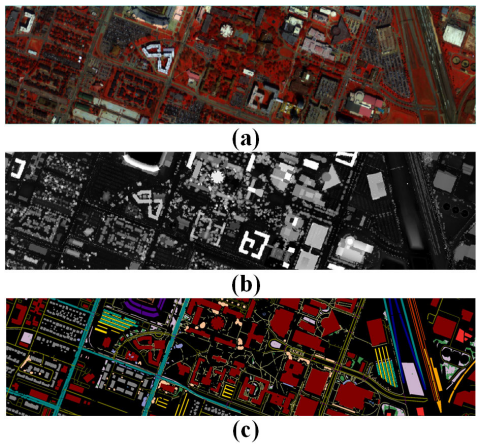


Fig. 11. Houston2018 dataset. (a) Pseudocolor composite image based on bands 50, 23, and 8 for HSIs. (b) Gray-scale image for the LiDAR-based DSM. (c) Ground truth.

but in practice, these regions also prominently contain substantial portions of "Forest." This discrepancy, i.e., the overlap between categories and the lack of correspondence with actual scene characteristics, poses a considerable challenge for model training. As a result, we restructure the dataset and refine the labeling process. The ground sample distance for all images is 2.2 m. Both HSI and LiDAR images have a spatial resolution of 1364 $\times$ 1636 pixels. The HSI data contain 101 spectral bands, spanning wavelengths from 400 to 1000 nm. This dataset encompasses six urban land cover categories, with detailed counts of training and test samples and corresponding category names provided in Table IV. Furthermore, Fig. 10 presents a pseudocolor composite of the hyperspectral data, a gray-scale LiDAR image, and the ground-truth map.

5) Houston2018 Dataset: Collected in February 2017 at the University of Houston campus, this dataset is provided by the 2018 IEEE GRSS Data Fusion Contest. The hyperspectral data are acquired using the ITRES CASI-1500 sensor, while the corresponding LiDAR data are captured by the Optech Titam Multispectral (MW) (14SEN/CON340) sensor. In this study, only a portion of the entire dataset is utilized. The dataset has a spatial resolution of 601 $\times$ 2385 pixels, with a ground sample distance of 1 m. Furthermore, the hyperspectral data consist of 48 spectral bands covering the wavelength range from 380 to 1050 nm. The dataset encompasses 20 urban land cover categories, with detailed counts of training and test samples, as well as the corresponding category names, provided in Table V. In addition, Fig. 11 presents a pseudocolor composite of the hyperspectral data, a gray-scale LiDAR image, and the ground-truth map.

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TABLE I

CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE HOUSTON2013 DATA. THE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Methods(Train/Test)	Coupled-CNNs	MAHiDFNet	ExViT	EndNet	FusAtNet	MFT	MDL	M2FNet	MACN	S^2ENet	Cross-HL	CAMFNet
OA（%）	87.45	86.30	92.19	83.55	78.68	88.42	87.50	91.72	93.30	93.01	94.22	**94.83**
AA（%）	89.67	88.41	93.45	84.59	81.87	90.35	89.40	92.92	94.40	94.20	95.21	**95.62**
K（%）	86.44	85.19	91.56	82.21	76.96	87.48	86.50	91.04	92.75	92.45	93.75	**94.41**
Healthy grass (20/1231)	97.08	97.89	97.97	89.52	95.78	96.91	94.64	95.86	98.05	**98.94**	97.97	96.83
Stressed grass (20/1234)	88.41	88.01	89.38	94.98	67.42	88.49	87.84	87.93	91.73	93.44	89.22	**97.08**
Synthetic grass (20/677)	99.11	99.11	98.23	99.85	**100.**	98.23	88.18	98.67	98.08	99.85	99.41	99.56
Tree (20/1224)	98.61	78.43	86.27	96.81	97.63	92.65	90.11	86.85	97.71	93.14	91.99	**98.77**
Soil (20/1222)	99.84	99.67	99.67	99.84	99.43	99.26	95.42	99.18	**100.**	**100.**	**100.**	99.51
Water (20/305)	99.34	98.36	98.03	96.39	98.03	98.03	**100.**	98.69	98.03	**100.**	98.69	**100.**
Residential (20/1248)	76.04	79.33	91.27	68.03	58.25	89.18	98.16	97.28	**99.68**	91.67	93.83	93.43
Commercial (20/1224)	70.34	71.41	79.49	74.10	81.86	86.85	80.96	79.66	66.18	77.53	87.01	**92.57**
Road (20/1232)	56.25	64.77	82.63	47.16	38.56	59.09	88.15	81.25	82.55	92.61	85.31	**94.16**
Highway (20/1207)	85.25	82.68	97.35	68.35	57.00	79.45	67.85	86.91	94.61	94.78	93.45	**97.76**
Railway (20/1215)	96.21	90.45	97.28	94.81	67.16	88.56	71.85	**100.**	99.18	**100.**	99.75	88.64
Parking lot 1(20/1213)	84.17	88.13	87.72	87.80	89.78	83.68	82.11	89.37	91.92	75.52	**93.32**	81.04
Parking lot 2(20/449)	94.43	87.97	96.44	51.45	89.53	94.88	95.77	92.20	98.22	95.55	**98.22**	95.55
Tennis court (20/408)	**100.**	**100.**	**100.**	99.75	89.46	**100.**	**100.**	**100.**	**100.**	**100.**	**100.**	99.75
Running track (20/640)	**100.**	**100.**	**100.**	**100.**	98.13	**100.**	**100.**	**100.**	**100.**	**100.**	**100.**	99.69

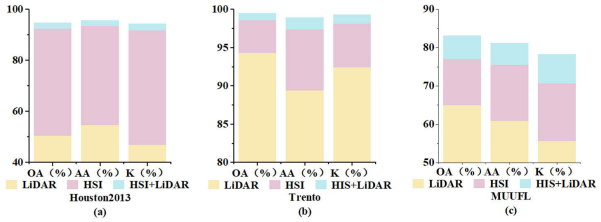


Fig. 12. Ablation analysis of different modal data inputs of the CAMFNet. (a) Houston2013. (b) Trento. (c) MUUFL.

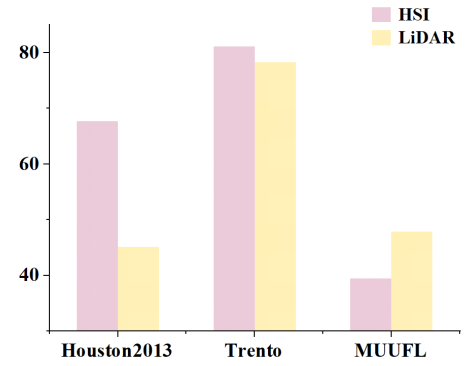


Fig. 13. Ablation analysis of different modal data inputs of ResNet.

B. Experimental Settings

1) Evaluation Metrics: In order to evaluate the classification performance of the CAMFNet and other state-of-the-art methods, four quantitative analysis metrics are computed, namely, overall accuracy (OA), average accuracy (AA), kappa coefficient (K), and accuracy per class. For each metric, higher values indicate a greater degree of accuracy in the classification process. The experiments are conducted on spectrally and spatially disjoint sets of training and testing samples.

2) Configurations: The experiments of the CAMFNet and other deep learning methods are conducted on the PyTorch platform using a server with an NVIDIA GeForce RTX 3080 Ti GPU for 100 epochs. The Adam optimizer is used for network optimization, with a batch size of 64.

C. Ablation Studies

1) Ablation Analysis of Various Modal Data Inputs: To understand how different modalities affect the classification performance of the model, we perform two sets of experiments using only HSI and LiDAR data inputs separately. The results are illustrated in Fig. 12. By evaluating the OA, AA, and K across the three datasets, the network consistently achieves better classification performance. This indicates that integrating both types of data significantly enhances classification outcomes and confirms that the proposed dual-branch network effectively utilizes the valuable information provided by these diverse modalities.

We also conduct unimodal classification tests using the ResNet model to assess the classification performance of individual modalities. As shown in Fig. 13, for the Houston2013 and Trento datasets, HSI outperforms LiDAR data in terms of classification accuracy. In contrast, on the MUUFL dataset, LiDAR demonstrates superior classification performance compared to HSI. Notably, our network consistently achieves superior classification results for HSI compared to LiDAR across three datasets, and these results exceed the corresponding unimodal classification performance of the ResNet model. This finding reveals two key insights: first, simple benchmark models may fail to effectively exploit the potential of HSI data; second, the contribution of each modality to overall classification performance varies across different scenarios. These observations further validate the feasibility of the AMF approach that we design.

2) Ablation Analysis of Different Components: To validate the effectiveness of the proposed modules, the contrastive learning strategy, the CMFIF module, and the CSI block, we conduct ablation studies on each module. Specifically, we test on three datasets using the same experimental setup, as shown in Table VI. INTER indicates SEFI and ESFI. An analysis of the table reveals that pretraining significantly enhances classification performance across all three datasets compared with the baseline model. This finding indicates that pretraining with a contrastive learning strategy strengthens multimodal feature representations by considering spatial distribution characteristics and semantic similarity.

For the Houston2013 dataset, excluding the CMFIF module leads to a significant decrease in the OA of 3.19%. For the

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TABLE II
CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE TRENTO DATA. THE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Methods(Train/Test)	Coupled-CNNs	MAHiDFNet	ExViT	EndNet	FusAtNet	MFT	MDL	M2FNet	MACN	S^2ENet	Cross-HL	CAMFNet
OA（%）	97.48	99.34	98.32	84.71	90.73	96.89	88.07	98.38	99.04	98.41	93.24	**99.49**
AA（%）	96.12	98.56	98.02	77.26	93.60	93.73	79.59	96.87	98.68	97.36	93.42	**98.94**
K（%）	96.64	99.11	97.76	79.38	87.94	95.85	84.12	97.84	98.73	97.88	91.10	**99.32**
Apple Tree (20/4014)	87.87	99.38	96.86	65.27	98.53	96.64	67.79	97.23	98.53	98.95	92.50	**99.53**
Building (20/2883)	96.98	96.39	95.77	71.73	**99.31**	**99.31**	56.47	93.62	97.26	92.44	92.54	98.16
Ground (20/459)	96.73	97.39	**98.47**	82.35	92.81	79.74	61.66	94.34	100.	98.26	93.68	97.60
Wood (20/9103)	99.96	**100.**	99.88	98.90	**100.**	**100.**	99.81	99.59	99.99	**100.**	99.99	**100.**
Vineyard (20/10481)	99.83	**99.97**	97.94	93.38	75.93	96.32	91.93	99.77	99.56	99.49	87.54	99.75
Roads (20/3154)	95.37	98.22	**99.21**	51.90	95.05	90.39	99.87	96.67	96.77	95.02	94.26	98.66

Trento dataset, the OA decreases by 0.51% under the same conditions, and for the MUUFL dataset, the OA decreases by 2.56%. These results demonstrate that the CMFIF module in shallow feature fusion effectively guides the extraction and fusion of spectral–spatial features and elevation features. In addition, the table presents the classification results of the network under separate INTER and AMF conditions, confirming the significant contribution of these interactions to feature acquisition when implemented individually.

Correspondingly, we verify the designed CSI block. Specifically, we substitute simple feature stacking for the CSI block and assess the final classification performance on the three different datasets, as detailed in Table VI. Compared with the baseline model, incorporating the CSI block results in improved classification outcomes on all three foundational datasets. The classification accuracy of the Houston2013 dataset increases by 3.82%, that of the Trento dataset increases by 0.46%, and that of the MUUFL dataset increases by 1.84%. This suggests that multimodal features' refusion is feasible in the deep feature fusion strategy.

Finally, CAMFNet delivers the best classification results across three fundamental datasets, which demonstrates the positive impact of the hierarchical fusion strategy on improving classification performance. Moreover, to further analyze the representational capability of the network learning objectives, we provide the feature visualization results before pretraining, after pretraining, and after fine-tuning the Trento dataset. The t-distributed stochastic neighbor embedding (t-SNE) plot of the features is illustrated in Fig. 14. It shows that the network demonstrates improved intraclass consistency and interclass separation after pretraining, with the fine-tuned network achieving the best performance overall. The results of this analysis serve to further validate the effectiveness of the CAMFNet and underscore the importance of each of its primary components.

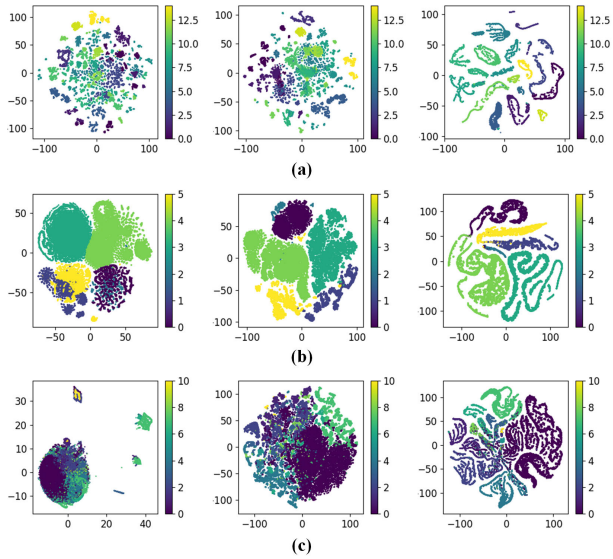


Fig. 14. Feature visualization on the three datasets. The first column represents features obtained without pretraining, the second column indicates features acquired through pretraining, and the last column illustrates the features of our final network. (a) Houston2013. (b) Trento. (c) MUUFL. Each point represents the features of a test sample, and its class label is indicated by a distinct color.

D. Classification Performance and Analysis

To evaluate the superior performance of the CAMFNet, several experiments are conducted on three datasets, comparing it with representative classification methods, including coupled-CNNs [17], multi-attentive hierarchical fusion net (MAHiDFNet) [19], extended vision transformer (ExViT) [51], EndNet [43], FusAtNet [42], multimodal fusion transformer (MFT) [52], multimodal deep learning (MDL) [27], multiscale 3-D–2-D mixed CNN feature extraction and multisource data lightweight attention-free fusion network (M2FNet) [53], mixing self-attention and convolution network (MACN) [54], spatial–spectral cross-modal enhancement network (S^2ENet) [55], and cross-hyperspectral and light detection and ranging (LiDAR) (HL) [56]. Specifically, coupled-CNNs propose a coupled network for multimodal information fusion, utilizing both feature-level fusion and decision-level fusion to represent heterogeneous features. MAHiDFNet employs a multilevel attention fusion strategy to integrate the modality-specific features and the modality-integrated features. ExViT employs parallel branches of position-shared transformers, augmented with separable convolution modules, to handle multimodal image patches. EndNet adopts an encoder–decoder framework that uses a reconstruction technique for feature fusion. FusAtNet implements a cross-attention mechanism that enhances the significance of HSI features by incorporating LiDAR data. MFT integrates optimal features from transformer encoders and multihead cross-patch attention. MDL utilizes weight sharing to facilitate interaction between different modalities. M2FNet uses a hybrid CNN feature extraction approach across multiple scales (3)-D–2-D) with the lightweight fusion of multimodal data. MACN combines adaptive convolutional encoding and a mixed self-attention convolutional Transformer to improve classification accuracy and computational

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TABLE III

CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE MUUFL DATA. THE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Methods(Train/Test)	Coupled-CNNs	MAHiDFNet	ExViT	EndNet	FusAtNet	MFT	MDL	M2FNet	MACN	S^2ENet	Cross-HL	CAMFNet
OA（%）	75.00	71.26	78.99	62.80	63.52	70.27	72.41	75.67	80.17	78.44	80.40	**83.24**
AA（%）	77.44	73.65	78.83	69.20	66.49	71.76	70.41	76.84	81.01	78.35	81.24	**81.24**
K（%）	68.46	64.17	73.15	55.15	55.27	63.08	65.23	69.34	74.84	72.33	75.05	**78.35**
Trees (20/23226)	80.41	72.02	85.08	58.37	67.12	70.31	78.17	78.80	81.08	**88.72**	82.61	88.64
Mostly grass (20/4250)	78.49	74.87	71.95	43.08	51.62	68.8	33.79	71.13	77.25	71.51	**80.73**	74.68
Mixed ground (20/6862)	43.72	49.24	60.84	49.69	35.88	42.07	51.71	63.51	63.76	**70.61**	64.27	62.61
Dirt and sand (20/1806)	**95.35**	81.56	93.52	60.74	65.67	92.97	81.73	91.03	89.87	85.60	94.68	84.77
Road (20/6667)	70.35	75.18	77.47	81.09	74.20	82.50	**92.79**	70.71	86.14	53.89	84.85	87.31
Wate (20/446)	**100.**	98.88	**100.**	98.43	99.10	99.78	99.55	98.88	**100.**	**100.**	**100.**	99.78
Building shadow (20/2213)	74.02	70.22	75.55	87.35	70.99	69.00	50.66	70.22	81.25	**87.93**	76.77	81.20
Building (20/6220)	91.88	89.53	85.69	79.74	75.85	87.32	90.14	85.63	91.24	77.27	87.25	**92.94**
Sidewalk (20/1365)	36.41	29.52	42.05	44.32	37.95	39.41	27.91	**55.16**	54.14	54.87	48.13	52.53
Yellow curb (20/163)	**81.60**	71.17	79.75	61.41	57.06	38.04	69.33	65.03	73.62	75.46	78.53	73.62
Cloth panels (20/249)	**99.60**	89.96	95.18	93.98	95.98	99.20	98.80	95.18	92.77	95.98	95.98	95.58

TABLE IV

CLASSIFICATION ACCURACIES (%) AND KAPPA COEFFICIENTS OF DIFFERENT MODELS ON THE AUGSBURG DATA. THE BEST ACCURACIES ARE SHOWN WITH THE BOLD TYPE FACE

Methods(Train/Test)	Coupled-CNNs	MAHiDFNet	ExViT	EndNet	FusAtNet	MFT	MDL	M2FNet	MACN	S^2ENet	Cross-HL	CAMFNet
OA（%）	84.25	94.70	87.82	56.02	88.57	85.86	46.99	87.82	94.15	94.58	91.12	**95.89**
AA（%）	87.01	93.29	87.73	61.95	84.07	81.13	54.40	88.51	94.03	91.99	90.55	**94.72**
K（%）	85.37	92.66	83.39	46.84	84.34	80.78	36.88	83.48	91.92	92.43	87.88	**94.28**
forest (20/4014)	88.31	96.50	86.04	40.00	90.29	85.49	23.29	85.47	93.29	**98.95**	88.92	97.23
grass (20/2883)	94.10	94.58	89.24	73.46	89.67	85.56	86.43	92.94	91.24	89.70	92.61	**94.55**
building (20/459)	93.29	90.96	88.72	56.49	95.90	97.35	60.13	89.49	**97.55**	86.32	97.27	94.35
water (20/9103)	97.29	96.08	95.02	95.00	93.29	98.36	91.80	91.47	97.22	95.88	**97.56**	96.90
road (20/10481)	79.98	87.97	72.64	88.47	37.43	47.00	25.24	83.77	90.10	85.84	74.68	**90.60**
railway (20/3154)	69.07	93.68	94.75	18.27	**97.86**	73.01	39.54	87.90	94.79	95.27	92.28	94.72

TABLE V

NUMBER OF TRAINING AND TESTING SAMPLES FOR THE HOUSTON2018 DATASET

NO.	Name	Train	Test
1	Healthy grass	1000	8799
2	Stressed grass	1000	31305
3	Artifcial turf	250	448
4	Evergreen trees	1000	12588
5	Deciduous trees	1000	4048
6	Bare earth	100	3516
7	Water	1000	166
8	Residential buildings	1000	38648
9	Non-residential buildings	1000	222460
10	Roads	1000	44737
11	Sidewalks	1000	32981
12	Crosswalks	1000	516
13	Major thoroughfares	1000	45318
14	Highways	1000	8850
15	Railways	1000	5937
16	Paved parking lots	1000	10475
17	Unpaved parking lots	100	49
18	Cars	1000	5587
19	Trains	1000	4375
20	Stadium seats	1000	5856

efficiency. S^2ENet leverages a spatial–spectral enhancement module with a forward-inverted CNN structure to effectively integrate hyperspectral and LiDAR data, achieving state-of-the-art performance in multimodal RS classification. Cross-HL introduces a novel self-attention module for multimodal RS data fusion, leveraging cross-hyperspectral and LiDAR attention to efficiently integrate complementary features for enhanced land use and land cover classification. These methods use the same training iterations and learning rates. To ensure a fair comparison, both training and testing datasets for these approaches are randomly selected.

1) Quantitative Results and Analysis: Tables I–IV present the OA, AA, K, and classwise accuracies of various methods on real-world RS datasets. By comparing these results, it can be observed that the proposed method significantly improves upon the other methods. As shown in Table I, on the Houston2013 dataset, the CAMFNet achieves the best performance in the OA, AA, and K, with values of 94.83%, 95.62%, and 94.41%, respectively. This illustrates that the proposed contrastive learning strategy successfully simulates the correlations and complementarities among multimodal samples.

The Houston2013 dataset exhibits discrete distribution characteristics, but CAMFNet still outperforms the other methods in most categories, achieving the highest accuracies in the stressed grass, water, tree, commercial, road, and highway categories. This notable improvement is attributed primarily to the numerous unlabeled regions within the Houston2013 dataset. Specifically, the dataset contains numerous instances of these classes that are not annotated, providing a rich source of unlabeled data. The CAMFNet leverages a contrastive learning strategy to effectively exploit these unlabeled data. By learning from the spatial distribution characteristics and semantic similarities between various samples, the CAMFNet captures comprehensive multimodal feature representations that enhance its performance.

Notably, CAMFNet is effective even for categories with similar spectral characteristics but differing elevation information, such as “Residential” and “Commercial.” Specifically, in the “Residential” and “Commercial” categories, the accuracy of CAMFNet consistently exceeds 92%. In particular, in the “Commercial” category, CAMFNet achieves an accuracy of 92.57%, far surpassing MDL (80.96%) and MFT (86.85%). Although S^2ENet, CAMFNet, and FusAtNet achieve the highest classification accuracy for healthy grass, stressed grass, and synthetic grass, respectively, CAMFNet overall provides the best classification results across all three types of grass, as shown in Fig. 15. These results underscore that CAMFNet enhances classification accuracy by using hierarchical fusion strategies for multimodal feature fusion.

We observe that in certain prediction tasks for minority classes, such as Parking Lot 1, the performance of our model

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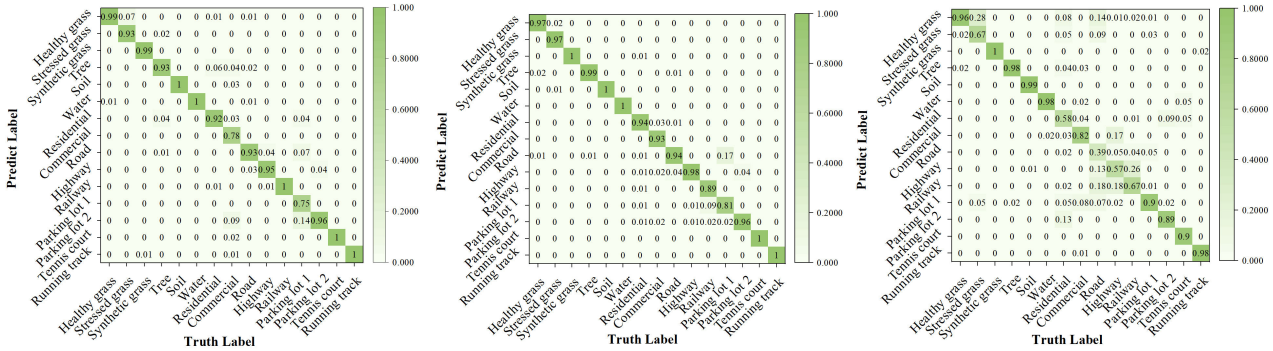


Fig. 15. Confusion matrix obtained for the proposed method on the Houston2013 dataset. (a) S^2ENet. (b) CAMFNet. (c) FusAtNet.

is suboptimal. This phenomenon may be attributed to the complex relationship between the spatial distribution and semantic similarity of positive and negative samples in contrastive learning. Specifically, when negative samples originate from regions that are spatially proximate, the model may receive misleading training signals, thereby impacting its performance. Although we have attempted to construct negative sample sets by randomly sampling a small number of unlabeled samples, this approach still struggles to fully mitigate the overlap of negative samples in spatial terms, particularly in scenarios such as Parking Lot 1, where the spatial distribution is highly concentrated.

In addition, as shown in Table II, on the Trento dataset, the proposed network exhibits advantages in the OA, AA, and K. Specifically, compared with the second-best method, CAMFNet yields improvements of 0.15% in OA, 0.26% in AA, and 0.21% in K. This indicates that CAMFNet's innovative design in multimodal data interaction and fusion effectively enhances complementary information from multiple modalities. We observe that the classification performance of the "Building" class in the Trento dataset is suboptimal, which contrasts with the classification performance for "Commercial" and other building-related classes in the Houston2013 dataset. This discrepancy may be attributed to the presence of a substantial number of unlabeled instances of the building-related classes in the Houston2013 dataset, which provides a rich source of unannotated data. CAMFNet capitalizes on a contrastive learning strategy to effectively leverage these unlabeled samples. By learning the spatial distribution characteristics and semantic similarities across various samples, CAMFNet captures a comprehensive multimodal feature representation, thereby enhancing its performance. In contrast, the Trento dataset contains relatively few unlabeled instances of the "Building" class. Comparable conclusions can be inferred from Tables III and IV regarding both the MUUFL and Augsburg datasets, further confirming the superiority of CAMFNet over comparative methods.

2) Visual Evaluation and Analysis: The visualization results of various comparative methods are presented in Figs. 16–19. These visualizations indicate that the CAMFNet method exhibits the highest similarity to the ground truth in the classification maps. Specifically, as shown in Fig. 16, in the Houston2013 dataset, due to the scattered distribution of various terrains and fuzzy boundary information, some classification methods exhibit more errors within the same class of ground objects. For example, certain railway areas are incorrectly classified into other categories, as highlighted by the red box. In contrast, CAMFNet fully considers the relevance and complementarity of hyperspectral and LiDAR data, achieving more precise classification results. Typically, for fewer pixels in discrete datasets, the proposed network achieves superior classification results.

As shown in Fig. 17, in the Trento dataset characterized by its uniform distribution, our proposed CAMFNet balances the spectra of each class better, resulting in more comprehensive classification outcomes and reducing errors in certain class pixels. For example, as illustrated by the red box, other methods result in significant misclassification in the Vineyard and Apple Tree regions, whereas our model produces smoother and more accurate classification results.

With respect to the MUUFL dataset, MAHiDFNet, EndNet and MDL exhibit noticeable noise and struggle to accurately identify object types, as shown in Fig. 18. By comparing some boundary regions, it is concluded that our method significantly outperforms other comparative methods in defining boundaries. Combining both the overall images and locally enlarged images shown in the red box, the proposed CAMFNet demonstrates effective performance even in scenarios with complex distributions of buildings. As shown in Fig. 19, similar trends are observed with the Augsberg dataset, where our model exhibits fewer classification errors compared to other methods. For example, at the boundary between forest and grass, the classification performance of the CAMFNet is significantly better than other models, as shown in the red box. This underscores the ability of the CAMFNet to integrate rich complementary information from HSI and LiDAR data under the guidance of contrastive learning and hierarchical fusion strategies.

3) Evaluation With Different Ratio of Training Samples: To assess the robustness and effectiveness of the CAMFNet, an analysis is conducted on how different sizes of training samples impact classification performance. For this study, a fixed quantity of labeled samples is randomly chosen for model training, with the remaining samples used for testing. Fig. 20 depicts the classification performance of all methods compared, varying by training sample size. Each class is assigned training samples of 5, 10, 20, 40, and 50. The results indicate that as the quantity of training samples rises, all methods exhibit enhanced classification performance, highlighting the reliance of deep learning models on data.

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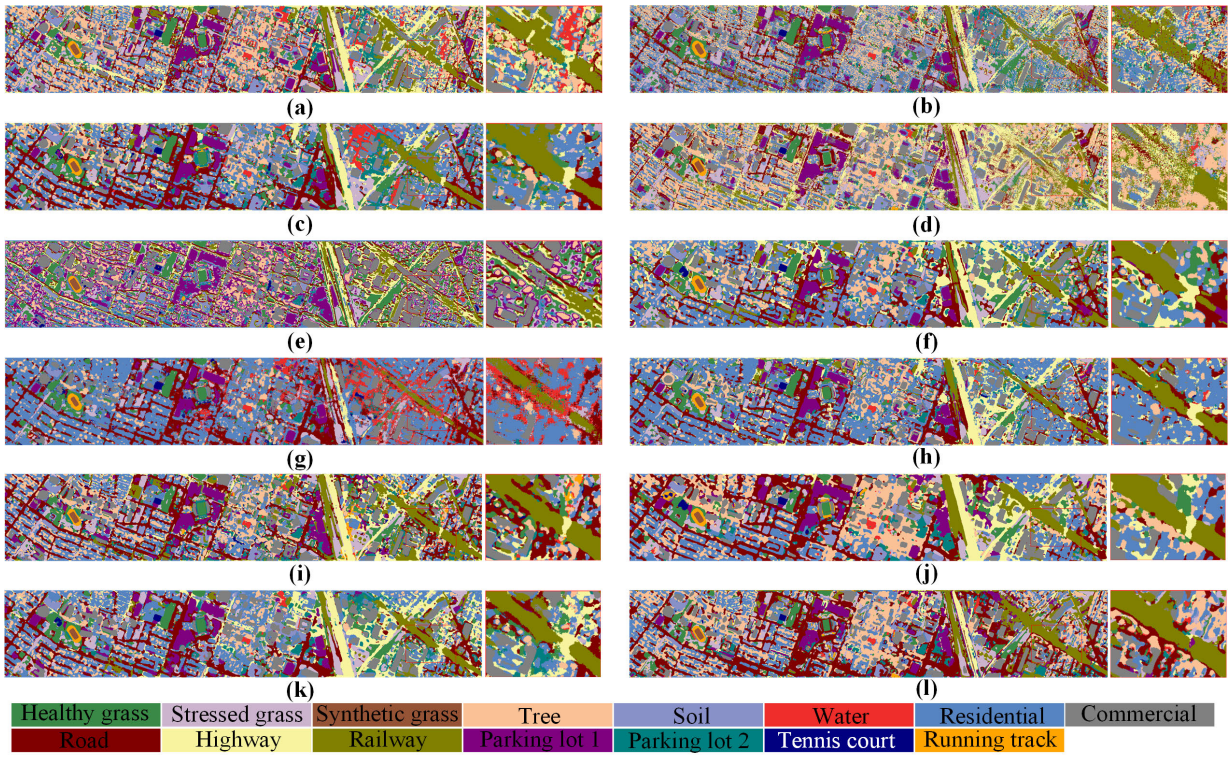


Fig. 16. Classification maps of the Houston2013 data using different models. (a) Coupled-CNNs. (b) MAHiDFNet. (c) ExViT. (d) EndNet. (e) FusAtNet. (f) MFT. (g) MDL. (h) M2FNet. (i) MACN. (j) S^2ENet. (k) Cross-HL. (l) CAMFNet.

Notably, CAMFNet generally outperforms other comparative methods under equivalent training sample sizes, which is particularly evident with fewer training samples. This advantage is attributed to the adoption of the contrastive learning strategy of MSE loss and SSC loss, which enables the model to extract potential complementary information across different modalities from unlabeled data during pretraining.

4) Analysis of Computational Complexity: Table VII illustrates the computational complexity of various methods across the Houston2013, Trento, and MUUFL datasets. Notably, the proposed CAMFNet ranks first in terms of parameters and second in floating-point operations (FLOPs), suggesting that its computational complexity surpasses that of most existing deep learning-based models. This can primarily be attributed to CAMFNet's adoption of a few-shot learning approach, which initially leverages contrastive learning for unsupervised feature extraction, capturing the latent information from multimodal features. Subsequently, the model undergoes fine-tuning with a small amount of labeled data. The architecture incorporates multiple modules to facilitate the interaction and fusion of multimodal features, thereby enhancing the model's representational capacity and feature extraction effectiveness, though this simultaneously increases the model's complexity. As a result, CAMFNet exhibits relatively high computational demands, particularly in terms of FLOPs. Nevertheless, CAMFNet significantly improves classification performance, with its accuracy notably exceeding that of other contemporary deep learning models. Furthermore, a comparison of inference times reveals that the performance of all models is comparable, with inference speeds remaining within the same order of magnitude. Hence, CAMFNet presents an efficient and robust solution for few-shot learning and multimodal learning tasks.

TABLE VI
ABLATION ANALYSIS OF DIFFERENT COMPONENTS

Pretrain	Baseline	INTER	AMF	CSI	OA (%)		
					Houston2013	Trento	MUUFL
	√				89.54	95.65	78.77
√	√				90.76	98.89	80.40
√	√	√			91.68	99.38	82.56
√	√		√		92.97	99.38	82.23
√	√	√	√		93.95	99.40	82.96
√	√			√	94.58	99.35	82.24
√	√	√	√	√	94.83	99.49	83.24

E. Parameter Tuning

1) Learning Rate: Fig. 21(a) illustrates the classification performance achieved by the proposed network across three datasets through variations in learning rates. When training multimodal RS image classification models, the learning rate directly impacts the model's convergence speed and effectiveness. A high learning rate can cause parameter updates to overshoot optimal values, leading to severe fluctuations in the loss function and hindering effective network convergence. Conversely, a low learning rate may trap the network in local minima, making it difficult to explore potentially better global solutions and, thereby, limiting model performance improvement. Therefore, appropriately adjusting the learning rate is crucial for ensuring stable convergence and optimizing the performance of deep learning models in multimodal RS image classification tasks. The network achieving optimal performance across all datasets when a batch size of 64 is used. The graph clearly shows that the optimal learning rate for all three datasets is 5e-3.

2) Batch Size: In the context of multimodal RS image classification, batch size serves as a crucial hyperparameter significantly influencing training efficiency, memory consump-

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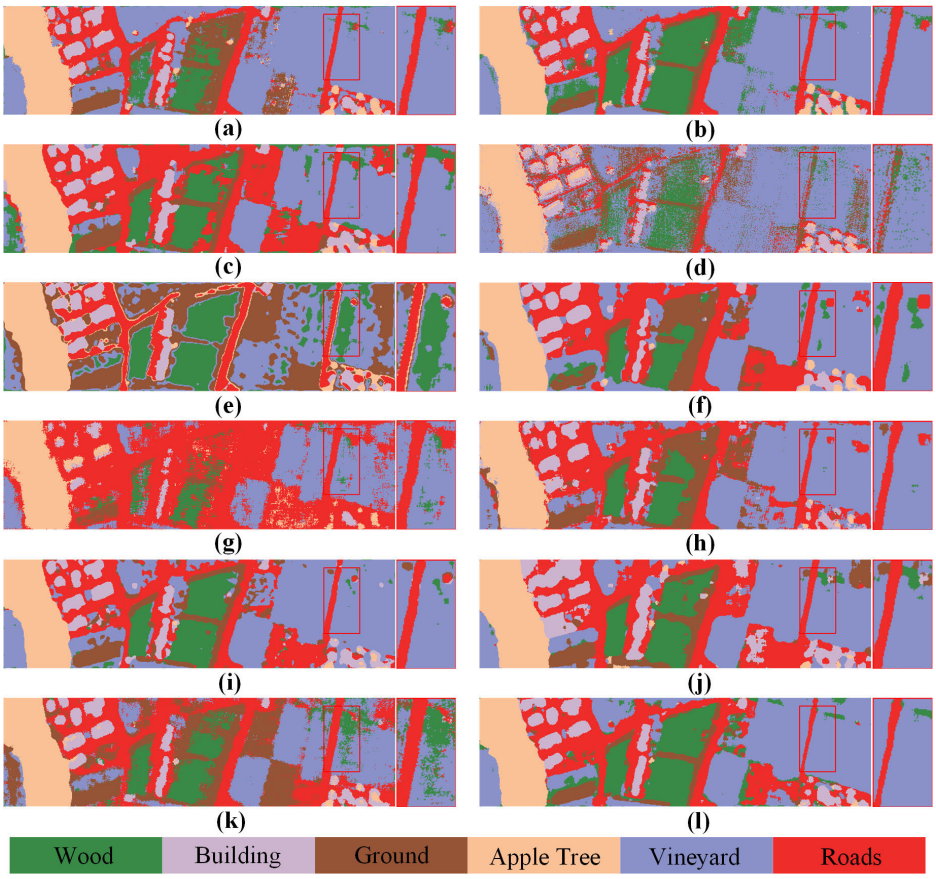


Fig. 17. Classification maps of the Trento data using different models. (a) Coupled-CNNs. (b) MAHiDFNet. (c) ExViT. (d) EndNet. (e) FusAtNet. (f) MFT. (g) MDL. (h) M2FNet. (i) MACN. (j) S^2ENet. (k) Cross-HL. (l) CAMFNet.

TABLE VII
COMPARISON OF THE NUMBER OF PARAMETERS AND FLOPS FOR DIFFERENT MODELS

Dataset	Complexity	Method											
		Coupled-CNNs	MAHiDFNet	ExViT	EndNet	FusAtNet	MFT	MDL	M2FNet	MACN	S^2ENet	Cross-HL	CAMFNet
Houston2013	Parameters	0.11G	15.32G	0.23G	0.21G	35.99G	0.23G	0.10G	99.70K	0.18G	0.14G	0.37G	**0.35M**
	FLOPs	0.17M	8.82M	2.09M	13.16G	**0.21T**	0.31M	0.33M	3.71M	0.68M	1.09M	0.19M	0.15T
	Test time	0.22s	0.71s	1.51s	0.21s	4.58s	0.32s	0.72s	5.64s	5.24s	0.70s	0.40s	**7.28s**
Trento	Parameters	0.10M	15.24M	0.23M	0.21M	35.98M	0.23M	0.10M	99.11K	0.18M	0.14M	0.37M	**0.35G**
	FLOPs	0.17G	8.81G	2.09G	13.12M	**0.21T**	0.31G	0.33G	3.71G	0.68G	1.09G	1.92G	0.15T
	Test time	0.41s	1.52s	2.81s	0.42s	9.11s	0.65s	1.44s	11.71s	11.40s	1.70s	0.80s	**16.44s**
MUUFL	Parameters	0.11M	15.28M	0.23M	0.21M	35.99M	0.23M	0.10M	0.10M	0.18M	0.14M	0.37M	**0.35G**
	FLOPs	0.17G	8.83G	2.09G	13.39M	**0.21T**	0.31G	0.33G	3.72G	0.68G	1.09G	1.92G	0.15T
	Test time	0.73s	2.52s	5.15s	0.71s	16.10s	1.13s	2.49s	22.63s	18.89s	2.40s	1.41s	**24.29s**

tion, classification accuracy, and generalization capability. Specifically, increasing the batch size accelerates the training process but increases the memory requirements. Conversely, reducing the batch size alleviates the memory burden but may lead to overfitting and stability issues. Determining the optimal batch size necessitates a comprehensive consideration of application requirements, dataset characteristics, and hardware conditions through systematic experimental evaluation. To evaluate the specific influence of batch size on model performance, we fix other hyperparameters and explore performance across the candidate set {8, 16, 32, 64, 128}. Fig. 21(b) clearly shows that the batch size notably affects the OA, with our proposed network achieving optimal performance across all datasets when a batch size of 64 is used.

3) Reduced Spectral Dimension: When applying principal component analysis (PCA) to process hyperspectral data, selecting an appropriate dimensionality reduction parameter q is a critical strategy to balance information retention, computational efficiency, and the suppression of noise and redundant information. Specifically, a smaller q improves

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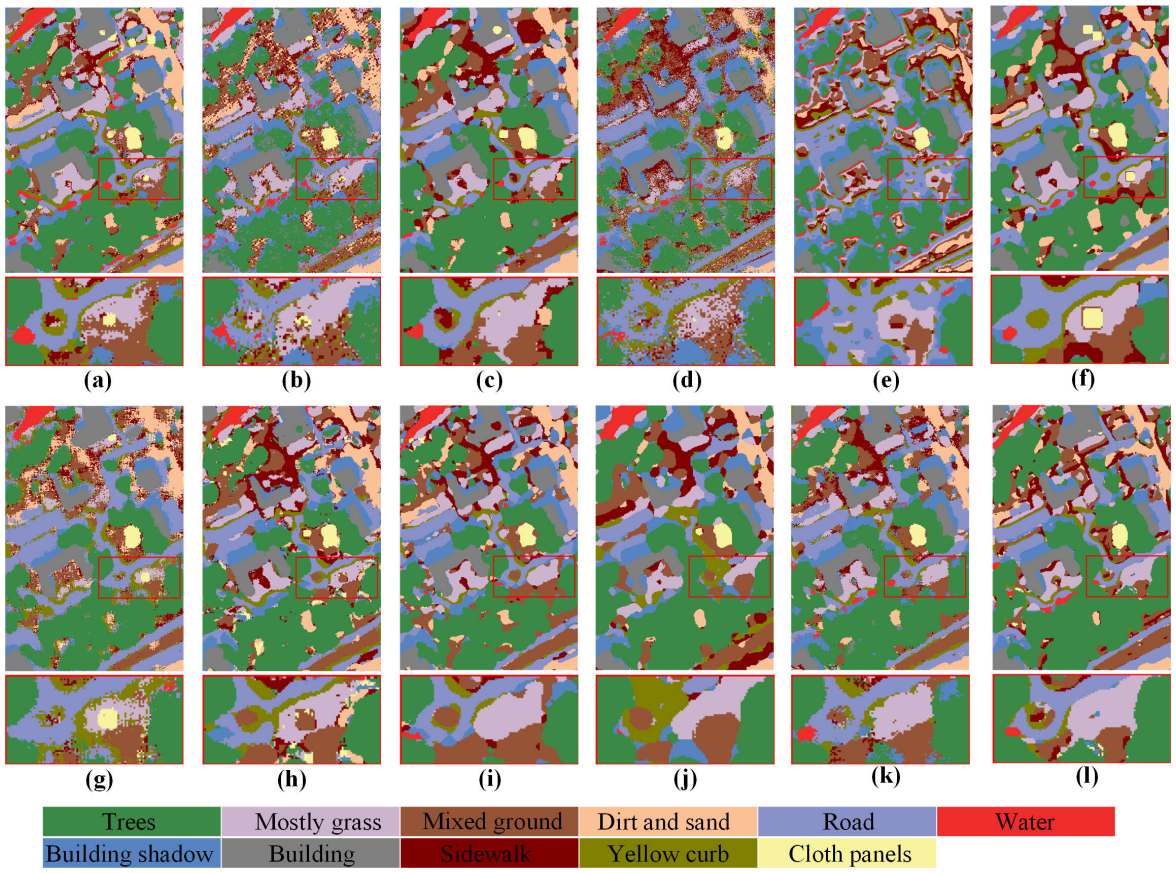


Fig. 18. Classification maps of the MUUFL data using different models. (a) Coupled-CNNs. (b) MAHiDFNet. (c) ExViT. (d) EndNet. (e) FusAtNet. (f) MFT. (g) MDL. (h) M2FNet. (i) MACN. (j) S^2ENet. (k) Cross-HL. (l) CAMFNet.

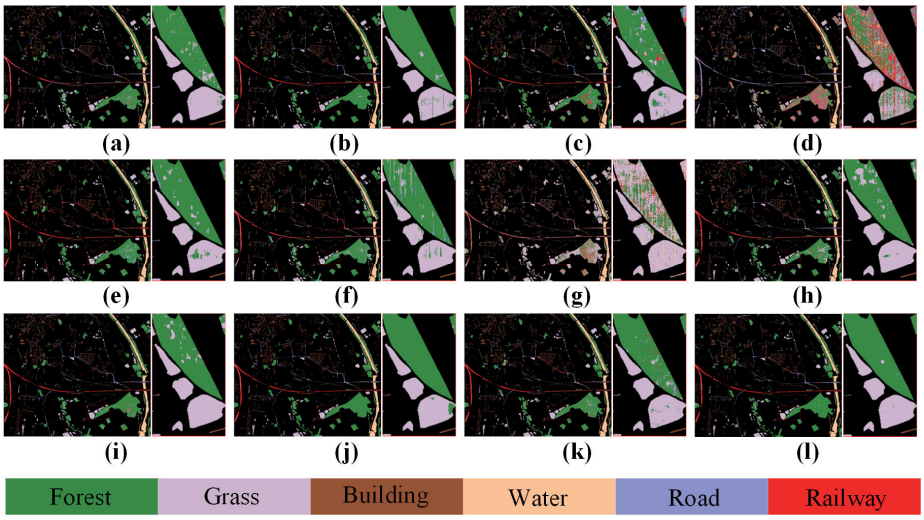


Fig. 19. Classification maps of the Augsburg data using different models. (a) Coupled-CNNs. (b) MAHiDFNet. (c) ExViT. (d) EndNet. (e) FusAtNet. (f) MFT. (g) MDL. (h) M2FNet. (i) MACN. (j) S^2ENet. (k) Cross-HL. (l) CAMFNet.

computational efficiency but leads to significant information loss, neglecting key features, and limiting classification performance. Conversely, increasing q preserves more useful information, initially increasing the classification accuracy. However, as q increases, redundant information and potential noise dominate, increasing model complexity and weakening generalizability, thereby increasing the risk of overfitting. To assess the impact of q, we maintain other hyperparameters constant and experiment with selected values from the predefined candidate set {1, 20, 30, 40, 50}. As shown in Fig. 21(c), we observe a trend of the OA across three different datasets, initially increasing and then decreasing. The CAMFNet attains the best results across all three datasets when q is set to 30.

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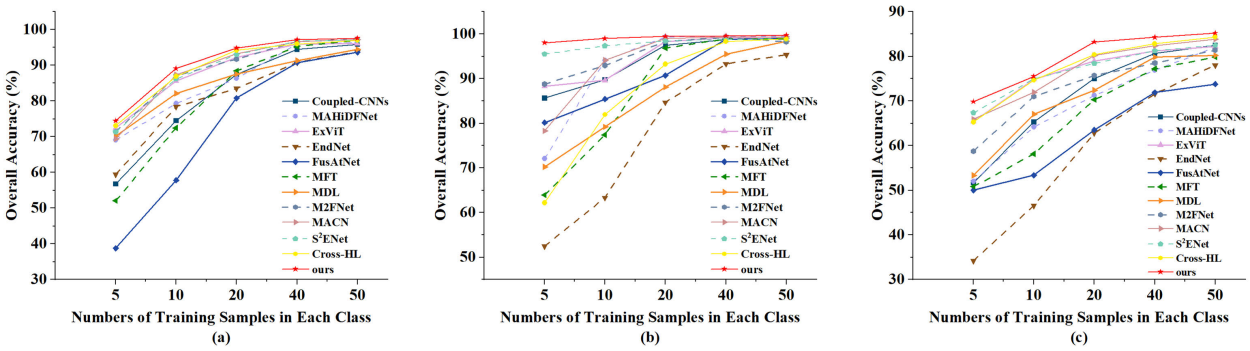


Fig. 20. Classification performance with different numbers of training samples. (a) Houston2013. (b) Trento. (c) MUUFL.

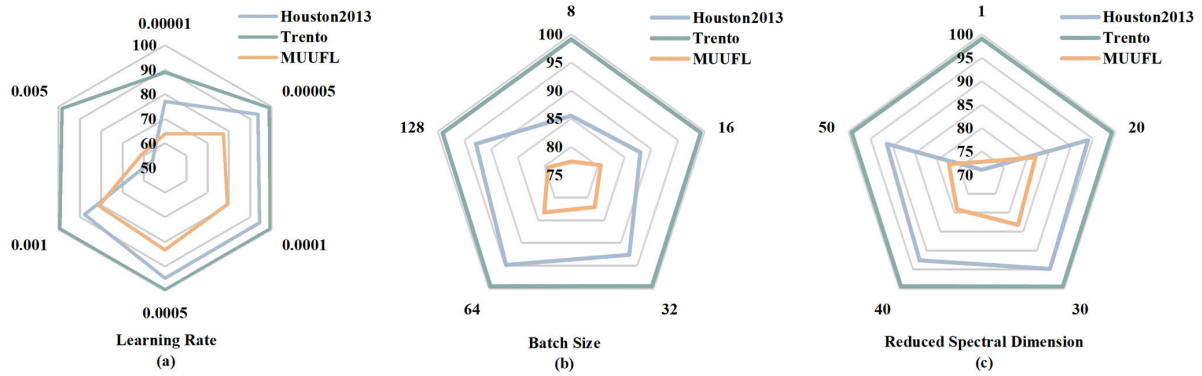


Fig. 21. Impact of different parameters on OA. (a) Learning rate. (b) Batch size. (c) Reduced spectral dimension.

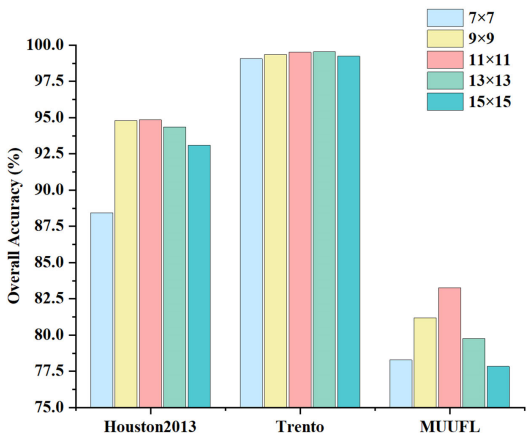


Fig. 22. Classification accuracy for different patch sizes.

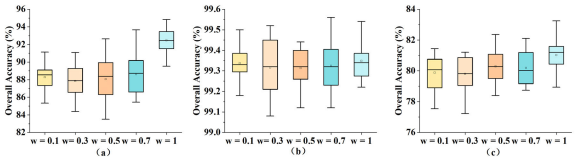


Fig. 23. Classification accuracy for different w values on the Houston2013 dataset. (a) Houston2013, (b) Trento, and (c) MUUFL.

4) Patch Size: In RS image classification, the size of the input patches significantly impacts the network performance. Larger patches offer a broad global perspective, allowing for deep exploration of spatial distribution patterns and interaction among features, thereby enhancing classification precision and accuracy. However, these methods involve increased information volume, leading to increased computational complexity and memory consumption, which limit processing efficiency and model scalability. Conversely, smaller patches focus on local details, accurately capture the internal characteristics of features, and are suitable for high-resolution identification. However, their limitation lies in their constrained field of view, potentially missing crucial spatial relationships and affecting overall classification consistency. Therefore, selecting an appropriate patch size is crucial.

We adjust the patch sizes to 7, 9, 11, 13, and 15 while keeping the other parameters at default values. We compare the classification results across three datasets, as depicted in Fig. 22. We observe that the OA of the methods initially increases steadily with increasing patch size until reaching an optimal peak, followed by a slight decline. This observation highlights the subtle influence of patch size selection on classification performance. Specifically, the optimal patch size for the Houston2013 and MUUFL datasets is 11, whereas for the Trento dataset, it is 13. This difference can be attributed to the unique geographical distribution characteristics of the Trento dataset, where extensive forest and agricultural regions necessitate a larger field of view to capture richer spatial context information, thereby improving classification accuracy. Notably, although larger patch sizes perform better on the

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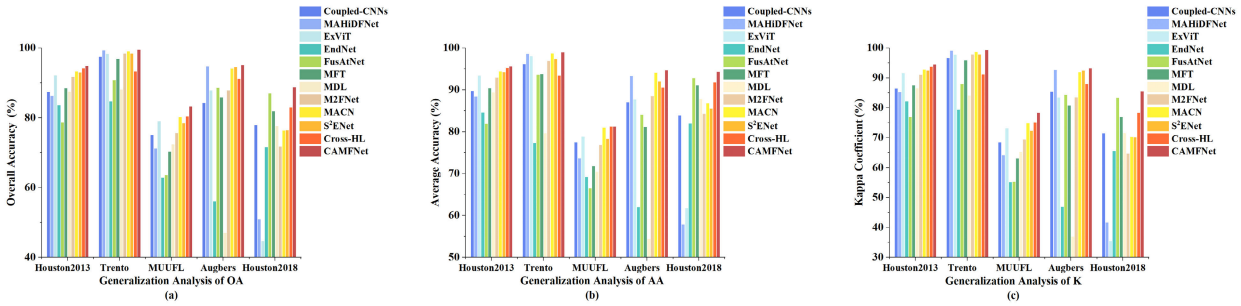


Fig. 24. Generalization analysis of different models on five datasets. (a) OA. (b) AA. (c) K.

Trento dataset, considering the overall performance across the Houston2013 and MUUFL datasets and maintaining the consistency and generality of the network structure, we ultimately decided to standardize the input patch size to 11.

5) Weighting Coefficient w: In contrastive learning, the introduction of the weighting coefficient w is crucial for balancing the MSE loss and SSC loss. By setting w reasonably, the impact of these two losses can be effectively coordinated to ensure that the model optimizes both feature alignment and the maximization of differences among negative samples. Specifically, a lower value of w places greater emphasis on the SSC loss, thereby enhancing the network's ability to capture spatial distribution characteristics and semantic similarities, which improves its generalizability in complex scenarios. However, an excessively low w may lead to insufficient focus on feature alignment, adversely affecting overall performance. Conversely, a higher value of w helps maintain the stability of feature alignment and reduces the risk of overfitting but may compromise the model's adaptability to complex scenes. To evaluate the optimal value of w, we conduct comprehensive experiments on the Houston2013, Trento, and MUUFL datasets, using candidates from the set $\{0.1, 0.3, 0.5, 0.7, 1\}$. As illustrated in Fig. 23, which corresponds to the growing weight of the MSE loss term in the total loss function, the classification accuracy exhibits a trend of initial decline followed by subsequent improvement. Specifically, when w is set to 1, the CAMFNet achieves the best classification results on all three datasets.

F. Generalization Analysis

To further investigate the generalization capability of our proposed CAMFNet, we extend the evaluation to include the Houston2018 dataset in addition to the Houston2013, Trento, MUUFL, and Augsburg datasets. This addition aims to assess the robustness and adaptability of our model across diverse RS scenarios. It is important to note that the Houston2018 dataset features pixel-level annotations with more granular categorization, such as distinguishing between residential buildings and nonresidential buildings, as well as sidewalks and crosswalks. Moreover, the dataset exhibits significant class imbalance. For example, the unpaved parking lots class contains only 149 instances, while nonresidential Buildings have as many as 223 460 labels. Consequently, using a contrastive model trained with merely 20 pixels per class fails to yield satisfactory classification performance. To address this, we increase the number of training samples per class.

As illustrated in Fig. 24, we comprehensively assess the generalization performance of our model across five major datasets using three evaluation metrics: OA, AA, and K. Remarkably, compared to other benchmark models, our model consistently demonstrates a robust leading position across all datasets and metrics. This outstanding performance not only underscores the model's strong generalization capability, ensuring its flexible adaptability to various datasets, but also further validates its high robustness and wide applicability across diverse RS scenarios.

V. Conclusion

In this article, a CAMFNet is proposed to address the joint classification problem of HSI and LiDAR. Specifically, CAMFNet uses semantic similarity contrast loss to promote information complementarity between different modalities and optimize the multimodal feature representation. In addition, a hierarchical fusion strategy is employed to integrate multimodal features. In the shallow feature fusion stage, the CMFIF module is proposed to promote the effective interaction of multimodal features and guide spectral–spatial and elevation feature extraction. Moreover, corresponding weights are dynamically assigned to different modal features to realize adaptive fusion. In the deep feature fusion stage, the CSI block is designed to refuse the multimodal features by using the negative feedback mechanism and make full use of the fused multimodal features to improve the fusion accuracy. Experiments on the Houston2013, Trento, MUUFL, Augsburg, and Houston2018 datasets demonstrate that CAMFNet outperforms various state-of-the-art methods. Although adaptive fusion can dynamically assign weights based on the importance of each modality, this may lead the model to focus more on feature extraction from the dominant modality while neglecting the suboptimal ones. Moreover, experiments on the Houston2018 dataset reveal that CAMFNet still faces considerable challenges in achieving strong performance on highly imbalanced datasets when only a small number of labeled samples are available. In the future, we will further explore more effective interaction and fusion of HSI and LiDAR data features in highly imbalanced scenarios, with the goal of ensuring the full utilization of each modality to achieve superior classification performance.

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